Resolution Geometry: The Classification of Information Loss

Contracts, Transported Composition, and Hidden Rooms

Matěj Rada
MatejRada@email.cz
Licensed under CC BY-NC-ND 4.0

October 2026

Abstract

RG's promise is a general, compositional rule that detects information loss and returns either all compatible objects or a justified abstention. This paper states that rule as a classification theorem relative to a declared *contract*. The contract fixes six things: the site, the family class, the gauge (punctured or through-wall), the protocol, the task type and an ordered test contract. Every earlier canary turns out to be a change of contract, which is why the contract must be explicit.

Within a contract the classification has five parts.

(i) *Pointwise:* exact residuals; postprocessing never cancels loss.

(ii) *Composition:* homotopy fibres compose by the transported Grothendieck construction and commute with germs.

(iii) *Framing:* unframed classes are framed rooms modulo link automorphisms.

(iv) *Stars (new general theorem):* for any compact Lie gauge acting definably, the rooms of an n-arm star are

$$\boxed{K_v \backslash \prod_{i=1}^{n} K_v / S_i^{\infty}}$$

where K_v is the stabilizer of the centre value and S_i^{∞} is the *limit stabilizer* of the i-th arm germ. The limit stabilizer is not the stabilizer of the limit value.

(v) *Isolated singular points:* two lifts are equivalent iff their transporter approaches a constant *uniformly* on the link; ray-wise limits do not suffice.

Globally, the obstruction tower of the print (L131) applies, and its germ steps are decided by (iv) and (v) wherever links are stars or punctured neighbourhoods lie in one stratum.

The theorem subsumes all earlier executed cases: sign germs, finite-group double cosets, coupled S_3 counts and the $U(1)$ torus. It adds a nonabelian compact case with continuous stabilizers, $SO(3)$ on $\mathbb{R}^3$, where rooms are $SO(3)\backslash(S^2)^n$ with an orientation that Gram matrices miss.

The out-of-contract boundaries are stated explicitly: noncompact gauges, non-tame families, groupoid-valued tasks, general stratified links, statistics and decidability.

Contents

1 What is being classified **2**

2 Contract semantics and certificates **3**

3 Pointwise loss **3**

4 Composition and germs **3**

5 Framed and unframed rooms **4**

6 Rooms at stars: the general compact theorem **4**

7 Isolated singular points **5**

8 Global classification **6**

9 Contract of the print's wall rungs (reconciliation) **7**

10 Outside every contract treated here **7**

11 Batteries **7**

1 What is being classified

The batteries leading here (I1, L2, S) exposed five facts. Each one becomes a requirement.

Fact	Requirement
Wall limits are incomplete: u and $\lvert u\rvert$ have equal limits and equal observations	Rooms are germ data, defined by restriction, not by limiting values (Section 6).
No universal finite jet order: r^{N+1} is a counterexample for every N	Finiteness is indexed by a declared complexity bound; otherwise germs are returned symbolically or the dispatcher abstains (Remark 6.7).
Punctured and through-wall gauges classify differently	The gauge class belongs to the contract (Definition 1.1).
A "prefix passed" flag is not proof	A lowest failure exists only for a certified prefix of a declared ordered test contract (Section 2).
The levels are test families, not one Postnikov tower	No global order is assumed; comparisons are proved maps.

Definition 1.1 (Contract)**.** A contract K fixes six items:

(i) a site of bases: compact definable sets in an o-minimal structure, with their open subsets and germs;

(ii) a family class: continuous definable maps $f : X \to V$;

(iii) a gauge: a compact Lie group G acting continuously and definably on V, together with the gauge class, punctured or through-wall continuous;

(iv) a protocol, i.e. an observation;

(v) the task type: set-valued and gauge-invariant (classified here), or groupoid- or bundle-valued (out of scope, Section 10);

(vi) an ordered test contract for dispatch.

The observation *separates orbits pointwise*: two values have equal observation iff they lie in one G-orbit. A generating set of invariant polynomials has this property for compact G [2].

2 Contract semantics and certificates

Imported Theorem 2.1 (Typed output; I1-D, S)**.** *A finite request returns exactly one of four outputs:*

- *DESCENDED, with a certificate covering the whole supplied domain;*
- *COMPATIBLE_SET, with every residual and an explicit witness pair;*
- *INCONSISTENT_MODEL, when no lift exists in the declared class;*
- *LEVEL_UNRESOLVED.*

A lowest failure is reported only when every earlier test in the declared order returned DESCENDED with a verified certificate. For incomparable tests the partial order of certified results is returned instead. Forged certificate fields and altered orders are rejected.

3 Pointwise loss

Imported Theorem 3.1 (Residuals and no cancellation; I1-A)**.** *For $F : X \to Y$, $G : Y \to Z$ and a task $a : X \to T$, put $R_F(y) = \{a(x) : F(x) = y\}$. Then*

$$R_{GF}(z) = \bigcup_{G(y)=z} R_F(y).$$

A task descends iff all residuals are singletons. If a fails to descend through F, it fails through GF: no postprocessing cancels loss. For several tasks the residual is the joint residual, not a product of marginal ones.

Imported Theorem 3.2 (Set-valued descent through groupoids; I1-C)**.** *For a functor $F : \mathcal{A} \to \mathcal{B}$ and an isomorphism-invariant set-valued task, the task factors through $\pi_0\mathcal{A} \to \pi_0\mathcal{B}$ on the essential image iff every residual over the homotopy fibre is a singleton.*

4 Composition and germs

Imported Theorem 4.1 (Transported composition; I1-B, S-T2)**.** *For $\mathcal{A} \xrightarrow{F} \mathcal{B} \xrightarrow{G} \mathcal{C}$ and $c \in \mathcal{C}$,*

$$\mathrm{hofib}_c(GF) \;\simeq\; \int^{h}_{(b,\beta)\in \mathrm{hofib}_c(G)} \mathrm{hofib}_b(F),$$

the Grothendieck construction with transport*. Its objects are $(a, b, \alpha : F(a) \to b, \beta : G(b) \to c)$, which map to $(a, \beta \circ G(\alpha))$. The intermediate arrow of a direct-fibre arrow u is forced: $v = \alpha' F(u)\alpha^{-1}$. Forgetting transport is wrong: for $* \to BC_2 \to *$, it would give two components instead of one.*

Imported Theorem 4.2 (Loss fibres commute with germs; S-T1)**.** *For groupoid-valued pseudofunctors on a filtered neighbourhood system,*

$$2\text{-}\operatorname{colim}_{p\in V} \mathrm{hofib}_{b|_V}(F_V) \simeq \mathrm{hofib}_{b_p}(F_p).$$

So germ-level loss of a composite is the composite of the germ-level losses. This statement provides no effective neighbourhood radius and no jet bound.

5 Framed and unframed rooms

Imported Theorem 5.1 (Framing; S-T3)**.** *For a restriction $R : \mathcal{A} \to \mathcal{B}$ (through-wall to punctured) and a link object b, the* framed room *is* $\mathrm{hofib}_b(R)$*. The unframed compatible classes are*

$$\{[a] : [R(a)] = [b]\} \cong \pi_0 \,\mathrm{hofib}_b(R)/\operatorname{Aut}_{\mathcal{B}}(b).$$

For two zero arms under S_3, the framed room has 6 components while the unframed room has 1.

The hidden room is what is lost by puncturing. Framed rooms are the object that composes. Unframed rooms are the output of classification.

6 Rooms at stars: the general compact theorem

Setting. Let X be a star: n closed arms $[0, \varepsilon)e_i$ glued at a centre 0. Let G be a compact Lie group acting continuously and definably on V. Let $f, h : X \to V$ be continuous definable families with equal arm-labelled pointwise orbits. A through-wall gauge is a continuous $g : X \to G$ with $h = g \cdot f$. For each arm put

$$T_i(r) = \{g : g f_i(r) = h_i(r)\}, \qquad S_i(r) = \mathrm{Stab}(f_i(r)) \qquad (r > 0).$$

Lemma 6.1 (Punctured transporters)**.** *After shrinking ε, each T_i has a continuous definable section on $(0, \varepsilon)$. Every continuous definable section has a limit as $r \to 0$.*

Proof. $T_i(r) \neq \varnothing$ because the orbits are equal. Definable choice gives a definable section, and the monotonicity theorem makes it continuous on a smaller interval. A definable curve in the compact set G has a limit [1]. □

Definition 6.2 (Limit stabilizer)**.** S_i^∞ is the set of limits at 0 of continuous definable sections of $S_i(\cdot)$.

Lemma 6.3. *S_i^∞ is a closed subgroup of $K_v = \mathrm{Stab}(v)$, where $v = f(0)$. It equals the upper Kuratowski limit of the family $S_i(r)$. If λ_i is the limit of any section of T_i, then the set of limits of sections of T_i is exactly the coset $\lambda_i S_i^\infty$.*

Proof. Subgroup. Products and inverses of sections are sections, and limits respect them.

Containment in K_v. If $s(r) f_i(r) = f_i(r)$, continuity of the action gives $s(0)v = v$.

Upper limit. Let p be an accumulation point of the definable set $\{(r, s) : s \in S_i(r)\}$ at $r = 0$. Curve selection gives a definable curve $(r(t), s(t)) \to (0, p)$ with r strictly monotone. Reparametrizing by r yields a section with limit p. The upper limit is closed, so S_i^∞ is closed.

Coset. If g_i is a section of T_i, then $t \mapsto g_i^{-1} t$ is a bijection between sections of T_i and sections of S_i. □

Theorem 6.4 (Star room theorem)**.** *Two lifts are equivalent by a through-wall gauge iff*

$$\bigcap_{i=1}^{n} \lambda_i S_i^\infty \neq \varnothing.$$

Hence, for a fixed observation with a reference lift f and centre value v, the unframed rooms are

$$\text{Room} \;\cong\; K_v \Big\backslash \prod_{i=1}^{n} K_v/S_i^\infty \;\cong\; S_1^\infty \Big\backslash \prod_{i=2}^{n} K_v/S_i^\infty,$$

with the diagonal action. A through-wall gauge exists iff a continuous one does iff a definable one does.

Proof. Necessity. A through-wall gauge restricts to sections of every T_i with the common limit $g(0)$.

Sufficiency. Take c in the intersection. On arm i write $c = \lambda_i s$ with $s \in S_i^\infty$, and use the section $g_i\sigma_i$ of T_i, where σ_i is a section of S_i with limit $\lambda_i^{-1}c$ (Lemma 6.3). Set $g(0) = c$. Arms are closed at 0, so the pasting lemma gives continuity. Furthermore $cv = \lim g_i(r)f_i(r) = h(0)$. The gauge built this way is definable.

Rooms. By the first part, $h \sim h'$ iff some c satisfies $\lambda_i' S_i^\infty = c\lambda_i S_i^\infty$ for all i. Every tuple of cosets with a common centre value is realized by the armwise constant lift $h_i = \lambda_i f_i$. Normalizing the centre to v leaves the residual group K_v, and $S_i^\infty \subseteq K_v$. □

Corollary 6.5 (Specializations)**.** *Theorem 6.4 specializes to all earlier executed cases:*

(a) Finite G: *S_i^∞ is the stabilizer of the entire arm germ. With two arms this gives the double coset $S_1\backslash G/S_2$ (L2-C). With more arms it gives the coupled quotient, e.g. 18 rather than 9 for S_3 (S-T4).*

(b) Sign germs: *this gives the count 2^B of L2-B.*

(c) Free $U(1)$ on nonzero values: *$S_i^\infty = 1$, and rooms form a $U(1)^{m-1}$-torsor (S-T5).*

(d) Zero arms: *$S_i^\infty = G$, and such arms contribute nothing.*

Remark 6.6 (What is new)*.*

- The theorem treats continuous, varying stabilizers. For $SO(3)$ acting on $\mathbb{R}^3$ with centre 0, S_i^∞ is the $SO(2)$ about the limit direction of arm i. Rooms are $SO(3)\backslash(S^2)^n$, of dimension $2n-3$: oriented congruence classes of the limit directions.
- The *limit* stabilizer, not the stabilizer of the limit value, is decisive. Using $\mathrm{Stab}(f(0)) = SO(3)$ would erase every room.
- With a nonzero centre, K_v is smaller, and variable gauges absorb first-order differences. Constant gauges cannot, so the two gauge classes differ.

Remark 6.7 (Determinacy, reconstruction and cardinality)*.*

- *Determinacy.* For polynomial arms the cosets $\lambda_i S_i^\infty$ are determined by the leading data at each arm's order of vanishing. There is therefore no universal jet order: r^{N+1} defeats every N. Under a declared degree bound d, a jet of order d suffices.
- *Reconstruction.* Gauges are explicit: h/f for free actions, minimal-rotation formulas for $SO(3)$.
- *Cardinality.* At bounded degree, room sets can still be continua: tori, or $SO(3)\backslash(S^2)^n$.

7 Isolated singular points

Theorem 7.1 (Uniform-limit criterion)**.** *Let a neighbourhood of x be a cone $C(L)$ over a compact definable link L. Suppose the punctured cone lies in one stratum of constant stabilizer type for f. Then $f \sim h$ through-wall near x iff some $c \in G$ satisfies*

$$\sup_{\ell \in L} \operatorname{dist}\big(c, T(\ell, r)\big) \longrightarrow 0 \qquad (r \to 0).$$

Proof. Necessity. A through-wall gauge is uniformly continuous on the compact cone, so it converges uniformly to $c = g(x)$.

Sufficiency. Fix a bi-invariant metric. On one stratum, T is a continuously varying family of cosets of conjugates of one closed subgroup. These are mutually isometric submanifolds and have a uniform normal injectivity radius ρ.

For small r the supremum is below ρ, so the nearest point $g(\ell, r) \in T(\ell, r)$ to c is unique and depends continuously on (ℓ, r). Moreover $\operatorname{dist}(g, c) \to 0$ uniformly. Setting $g(x) = c$ gives a through-wall gauge. □

Corollary 7.2 (Functional moduli)**.** *For free actions the transporter is a single point $t(\ell, r)$. Two lifts h, h' are equivalent iff $t'\, t^{-1}$ tends uniformly to a constant. When t converges uniformly to a function $\tau : L \to G$, rooms are such limit functions modulo left constants. This is the functional modulus found at codimension-2 collisions.*

No-Go 7.3 (Ray-wise limits are insufficient). Take $L = S^1$, free $U(1)$ on nonzero complex values, and $h = e^{i\pi b} f$ with $b = xy^2/(x^2 + y^4)$. Along every ray the gauge tends to 1. On the curve $x = y^2$, however, it equals i at every radius, so no through-wall gauge exists.

A link therefore cannot be treated as a family of independent arms. Nor can it be replaced by one fixed-radius link without a comparison theorem.

8 Global classification

The print's obstruction tower (L131) gives the global criterion: isomorphism holds iff the transporter has a section, iff every germ step and every cell class in $[S^{r-1}, H_\tau]$ can be passed. Sections 6 and 7 decide the germ steps at two kinds of point:

- points whose links are stars (Theorem 6.4);
- points whose punctured neighbourhoods lie in one stratum (Theorem 7.1).

At a point whose link meets several strata, the germ step is again a transporter-section problem, now on $L \times (0, \varepsilon)$ with a uniform limit, over a base of lower dimension. We state this reduction. We do not prove that it yields a computable criterion in general.

Main Theorem (Classification of information loss within a contract). *Fix a contract* K *with compact Lie gauge, tame continuous families, a pointwise orbit-separating observation, set-valued invariant tasks and through-wall continuous gauges.*

1. *A task descends iff its residuals are singletons. Composite residuals are unions. Loss is never cancelled by postprocessing.*
2. *Loss objects compose by the transported Grothendieck construction and commute with germs.*
3. *Unframed classes are framed rooms modulo link automorphisms.*
4. *At a star, the rooms are $K_v \backslash \prod K_v / S_i^\infty$.*
5. *At an isolated singular point, equivalence holds iff the transporter has a uniform constant limit.*
6. *Globally, isomorphism holds iff the obstruction tower vanishes, with germ steps decided by 4 and 5 wherever they apply.*

The dispatcher returns one of four typed outputs (descended; compatible set with witnesses; inconsistent model; level unresolved), together with certificates. It reports a lowest failure only for a certified prefix.

9 Contract of the print's wall rungs (reconciliation)

L129–L131 of *Resolution Geometry* are stated in the contract:

- C^0-definable families over compact definable bases;
- gauge $G = O(p) \times O(q)$, compact;
- through-wall continuous gauges, which by Theorem 6.4 is equivalent to definable gauges;
- unframed set-valued classification.

They are consistent with the boundaries listed in Section 1:

- L129's rooms are limits of *frames*, i.e. the cosets $\lambda_i S_i^\infty$, not limits of values.
- L130's no-go is the functional modulus of Theorem 7.1.
- L131's germ step is not a finite-jet condition.

The print should declare this contract explicitly; that erratum is not yet applied.

Two claims of the earlier programme draft are withdrawn:

- the blanket use of Brown's algorithm;
- "nonabelian π_1 implies undecidability".

Both require explicit input models and reductions.

10 Outside every contract treated here

1. *Noncompact gauges* (GL, Lorentz). Limits may diverge, and Lemma 6.1 fails.
2. *Non-tame families.* An example is Rellich's oscillating eigenvectors.
3. *Groupoid- or bundle-valued tasks and higher stacks.* These need coherent descent data; only set-valued tasks are classified.
4. *General stratified links.* The reduction is stated, not proved computable.
5. *Statistical identification* from samples, beyond declared interval certificates.
6. *Decidability* beyond finite, effective inputs.

11 Batteries

Battery	Content
I1 (imported)	Residual composition (960 composites, 12 432 task contracts, none cancelled); transporter equivalences (cyclic and S_3); typed probes; 7 canaries
L2 (imported)	Polynomial sign germs (2 430 families); count 2^B; S_3 double cosets (100 contracts, 165 rooms); jet-cap canary r^{N+1}; punctured versus through-wall canary
S (imported)	Certificate dispatcher (340 tables); 1 110 S_3 multi-arm contracts; 559 872 hom-set checks; framed versus unframed (6 versus 1); $U(1)$ (1 296 pairs)

Battery	Content
`rg_room_general.py` (new)	R1: $SO(3)$ stars, 60 contracts, prediction by limit cosets exact, explicit gauges (residual $< 10^{-12}$), chirality canary. R2: limit versus value stabilizer. R3: nonzero centre, variable versus constant gauge. R4: specializations (18 versus 9; torus). R5: ray-limit no-go.

Limits. All examples are synthetic. The new theorems rest on o-minimal geometry (definable choice, monotonicity, curve selection) and on compact Lie theory, which are cited, not reproved. Sampling cannot prove theorems; it can only witness them or find counterexamples.

References

[1] L. van den Dries, *Tame Topology and o-minimal Structures*, Cambridge University Press, 1998.

[2] A. L. Onishchik, E. B. Vinberg, *Lie Groups and Algebraic Groups*, Springer, 1990 (invariants of compact groups separate orbits).

[3] G. E. Bredon, *Introduction to Compact Transformation Groups*, Academic Press, 1972.

[4] A. Grothendieck, *Revêtements étales et groupe fondamental* (SGA 1), Exposé VI, Springer LNM 224, 1971.

[5] M. Rada, *Resolution Geometry* (print edition, L0–L133), 2026.

[6] M. Rada, *Identification programme I1* (`rg_identification_I1.py`), 2026.

[7] M. Rada, *L2 battery: polynomial germs and finite-group rooms* (`rg_L2.py`), 2026.

[8] M. Rada, *S battery: germ composition and room classification* (`rg_S_germ.py`, `rg_S_U1.py`), 2026.